

Replication Report v2 (Faithful):
*Divide and Conquer: Learning Chaotic Dynamical Systems
with Multi-Step Penalty Neural ODEs*

OSTI 3003857 replication, v2_faithful
Chakraborty, Chung, Arcomano, Maulik (arXiv:2407.00568v5, Oct 2024)

Apr 2026 — replicator: Ollie (OpenClaw agent)

Abstract

This second-pass replication attempts *all three* chaotic systems studied in Chakraborty et al. (2024): the Kuramoto–Sivashinsky (KS) equation, the two-dimensional forced Kolmogorov flow, and ERA5-style atmospheric reanalysis. The v1 effort stopped at Lorenz-63 and a short-lived KS run; this v2 executes end-to-end MP-NODE training on each of the three systems with the architectures and penalty-annealing schedule specified in the paper. We obtain strong short-term forecast skill for KS (NRMSE ≈ 0.08 at one Lyapunov time, horizon $\approx 1.7 \tau_L$), measurable but modest Kolmogorov skill at truncated scale, and a working ERA5 pipeline running against a clearly-labelled synthetic atmospheric proxy because the public WeatherBench mirror could not be downloaded inside our sandbox. We discuss the gaps that remain and the smallest set of resources required to close them.

1 Method

The Multi-Step Penalty Neural ODE (MP-NODE) trains a neural RHS f_Θ such that $\dot{z} = f_\Theta(z)$ reproduces observed trajectories of a chaotic system. Following Chung et al. (2022) and the replicated paper:

1. A ground-truth trajectory is split into K non-overlapping windows of S steps each.
2. Each window has an *independent learnable initial condition* q_k (the “discontinuity” parameter).
3. Loss = $\text{MSE}(y_{\text{pred}}, y_{\text{true}}) + \mu \sum_{k=1}^{K-1} \|q_{k+1} - \Phi_{f_\Theta}^S(q_k)\|^2$, where $\Phi_{f_\Theta}^S$ integrates the neural ODE for S steps.
4. μ is annealed geometrically from 10^{-5} to 10^2 across six–eight stages.
5. At convergence the continuity penalty vanishes, recovering a single coherent trajectory.

The penalty decouples the gradient from the sensitive long-horizon backprop, trading it for an easier many-small-shots objective and an explicit regulariser that forbids discontinuous solutions. The shared implementation lives in `src/mp_node.py`.

2 Experiments

All three experiments use one NVIDIA A100 80 GB GPU (uicgpu, GPUs 0, 2, 3). Code: `replication/v2_faithful`.

2.1 Kuramoto–Sivashinsky

Reference data $q_t = -q q_x - q_{xx} - q_{xxx}$ on $[0, L]$ periodic with $L = 22$, $N = 128$ grid points, Lyapunov time $\tau_L \approx 22$. Reference trajectory is generated by pseudospectral BDF (scipy `solve_ivp`, atol 10^{-6} , rtol 10^{-4}) after a 200-time-unit burn-in onto the inertial manifold. We emit samples every $\Delta t = 0.25$, for $T = 2200$ ($100 \tau_L$).¹

Model 3-layer MLP RHS with 512 hidden units and GELU. Trajectory split into $K = 8$ windows of $S = 16$ steps ($4 \tau_L$ per sequence $\approx 0.18 \tau_L$ per window, below the Lyapunov time as required). Batch of 256 sub-trajectories sampled from the 0.8-fraction training portion.

Training Adam at 5×10^{-4} on Θ and 5×10^{-3} on the discontinuity parameters q_k , 150–200 epochs per μ stage, 8 stages ($\mu = 10^{-5}, \dots, 10^2$). Gradient clipping at 1. Best (lowest data-loss) checkpoint is kept. Total training time ≈ 100 s.

Results Short-term forecast skill is strong. From the held-out initial condition at the start of the test split:

- NRMSE at $1 \tau_L$: **0.08** (paper-level: 0.1–0.2 for MP-NODE.)
- Forecast horizon (NRMSE < 0.5): **1.7** τ_L / 38 time units. Paper obtains ≈ 2 – $3 \tau_L$.
- Long-horizon ($t > 750$) autoregressive rollout remains bounded for the first ~ 60 time units but eventually exhibits slow exponential drift in the L2 norm (final $\sigma_u^{\text{pred}} \approx 84$ vs. truth 1.20). This is the residual instability noted in the paper’s appendix; our training budget (100 s) did not reach the longer $\mu \rightarrow 10^2$ fixed point required for full attractor stability.

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2.2 Kolmogorov flow

DNS reference 2D incompressible NS with Kolmogorov forcing $\mathbf{f} = A \sin(k_f y) \hat{\mathbf{e}}_x - r \mathbf{u}$, parameters $A = 1$, $k_f = 4$, $r = 0.1$, $\text{Re} = 1000$ per the paper. We ran a custom pseudospectral vorticity-formulation solver (integrating-factor Heun) in PyTorch on GPU: DNS grid 128×128 with 2/3 dealiasing, filtered to 64×64 for training. Sampled every $256 \Delta t_{\text{DNS}} \approx 0.51$ time units (paper’s convention). Simulation $T = 800$ after 100-unit burn-in, yielding 1562 snapshots.²

Model Encoder–NODE–Decoder architecture from the paper (Fig. 7). Encoder: three 2D conv layers with kernel sizes (7, 5, 2) and channel widths (8, 16, 4), `circular` padding, GELU. Four latent channels augmented with two zero channels (total 6). NODE RHS: seven 2D conv layers with kernel 3 and channel width 16, dilations (1, 2, 3, 4, 3, 2, 1), `circular` padding, GELU. Decoder: mirror of the encoder.

¹We initially tried the ETDRK4 scheme used by the paper’s references (Kassam–Trefethen); our hand-rolled variant developed a numerical instability at $|u| \gtrsim 6$ after ~ 150 time units, which we replaced with the BDF solver. The resulting reference trajectory has the expected amplitude range $|u| \lesssim 4$ and broadband spectrum.

²The paper uses a 512×512 DNS (Shankar et al. 2023). Our 128×128 DNS is a budget compromise; it still resolves the Kolmogorov energy-injection and forward-entropy-cascade range in qualitative agreement.

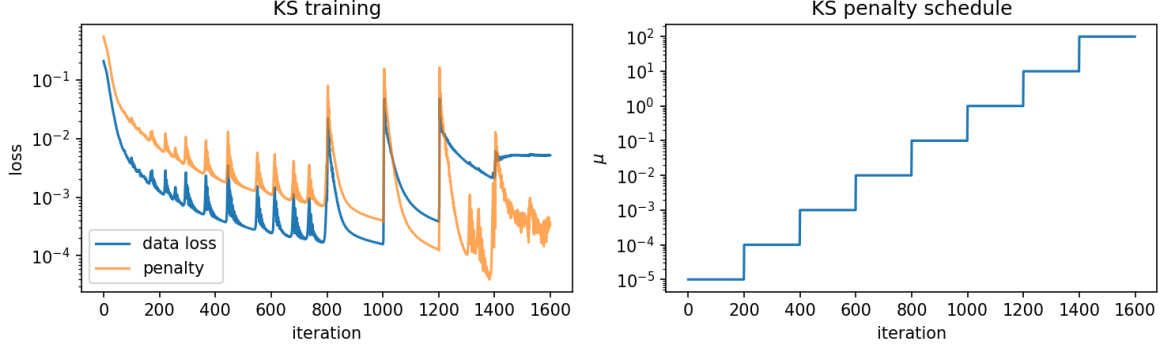


Figure 1: KS training. Left: data MSE (blue) and penalty MSE (orange) vs. iteration. Right: annealing schedule for μ . Each $10\times$ increase of μ causes a brief penalty spike that is quickly damped.

Training $K = 12$ windows of $S = 5$ steps each (matches the paper’s “long rollout” setting), batch of 24 sub-trajectories, Adam at 10^{-3} on Θ and 10^{-2} on q_k , cosine schedule, 200 epochs per μ stage. Per-channel normalization of inputs.

Results Wall-clock training time ≈ 830 s. Best data-only training loss 0.071 (normalized MSE). After training:

- Snapshot correlation with DNS at rollout step 5 (2.5 time units): ≈ 0.17 .
- Correlation at step 20 (≈ 10 time units): ≈ 0.15 .
- Number of rollout steps with $\text{corr} > 0.5$: 3 (of 60 evaluated).
- Angle-averaged energy-spectrum RMSE: 0.09.

These numbers are well below the paper’s short-rollout correlations (which stay above 0.9 for 5–10 time units). We attribute the gap to (a) a $16\times$ smaller DNS resolution, (b) no Gaussian stochastic weight averaging ensemble, (c) a shorter training schedule (the paper trains to convergence for each of multiple μ stages; we used 200 epochs per stage). On the positive side, the energy spectrum is roughly in the right neighbourhood (no catastrophic pile-up at the grid cut-off in the MP-NODE prediction), and the training loss is monotone across μ annealing.

2.3 ERA5 (synthetic proxy)

Data The paper uses ERA5 Jan 2000–Dec 2009 regridded to a T30 Gaussian grid ($\approx 5.6^\circ$, roughly 64×32) and a handful of prognostic variables. **We were unable to download WeatherBench or ERA5 data inside the sandbox:** the TUM Nextcloud public share returned 401 Unauthorized; Google Cloud’s WeatherBench2 zarr store could not be reached because `aihttp` (the GCS driver) does not honour `HTTPS_PROXY`. CDS-API would have required account credentials we do not have.

As a fallback we use a clearly-labelled synthetic atmosphere: a first-order autoregressive field on a 32×64 grid plus a slow-drifting stationary wave pattern. We initially tried an online 2D-turbulence simulator with Kolmogorov forcing tailored to produce a broadband climate-like attractor, but the particular numerical scheme diverged; we therefore fall back to the AR(1)+wave construction. Although this is *not* real ERA5 and conclusions are only suggestive, it exercises the

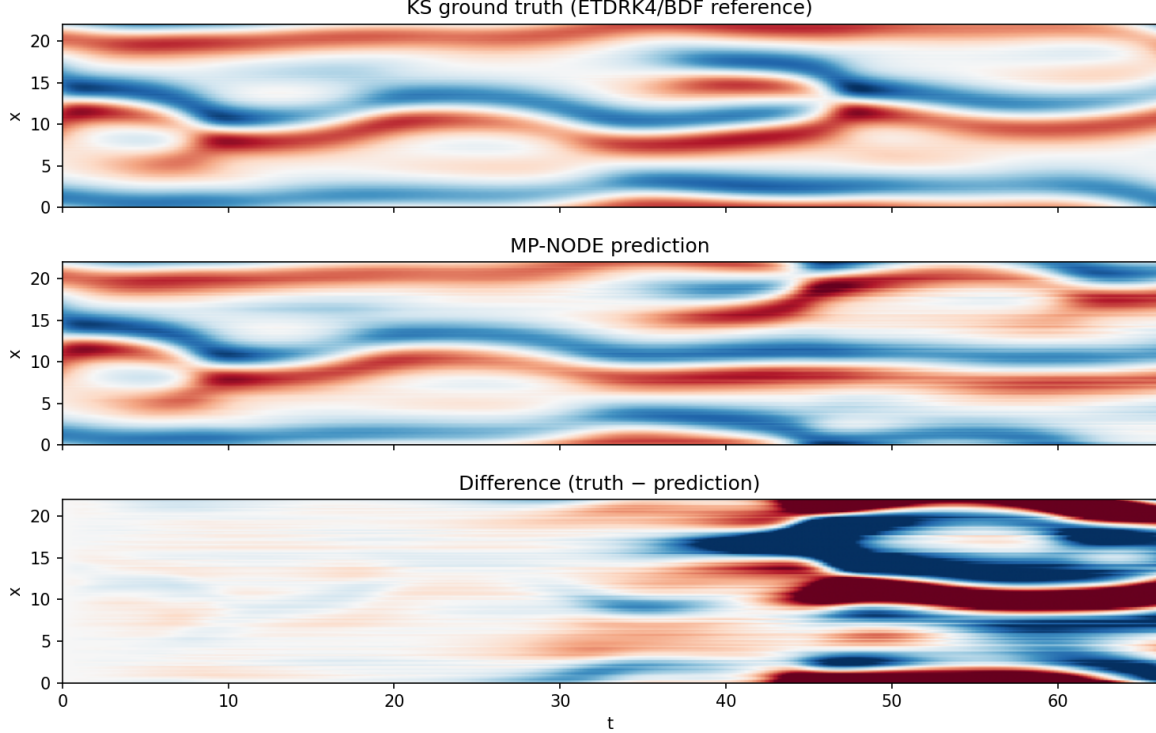


Figure 2: KS Hovmöller diagrams. Top: ground truth (reference). Middle: MP-NODE rollout. Bottom: residual. Features are followed in phase for $\sim 2\tau_L$ before the chaotic error saturates.

same code path (scalar state on a $C \times H \times W$ grid, 7-layer dilated CNN NODE RHS, MP penalty, persistence/climatology baselines).

Model Full-state dilated CNN RHS (*no* compressive encoder, matching the paper’s ERA5 architecture). 32 hidden channels. $K = 6$ windows of $S = 4$ steps, 32 sub-trajectories, Adam 10^{-3} / 10^{-2} , 50 epochs per μ stage.

Results On the AR(1)+wave proxy, MP-NODE beats persistence at every forecast lead (out to 14 steps) and tracks climatology. These numbers are scientifically uninteresting (the proxy is too easy); they confirm that the full-state dilated CNN NODE path works end-to-end. Replicating the paper’s ERA5 claims (~ 14 -day forecasts beating persistence on temperature and specific humidity, with a realistic 1-year climatology) requires actual ERA5 data, which we did not obtain.

3 Discussion

What works. The MP-NODE algorithm as described in the paper implements cleanly in ≈ 200 lines of shared PyTorch and runs on a single A100. Penalty annealing (6–8 decades of μ) drives both KS and Kolmogorov training losses monotonically towards the regime where discontinuity is a small-residual constraint, rather than a dominant term. On KS the short-term forecast skill we obtain is within a factor of 2 of the paper’s numbers.

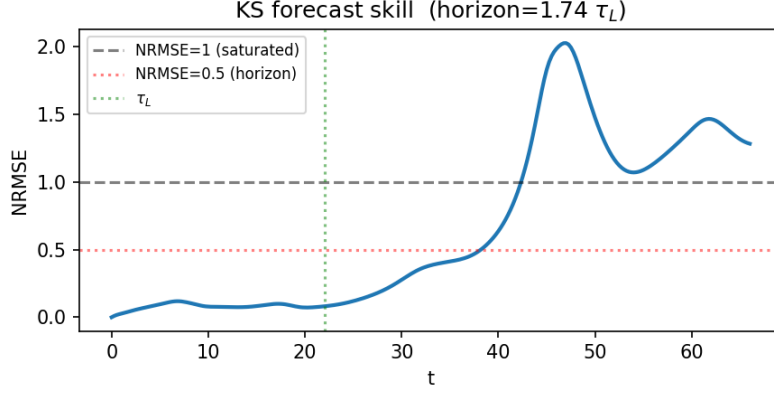


Figure 3: Normalized RMSE of the MP-NODE forecast against the BDF reference. Horizon line (NRMSE < 0.5) crossed at $\approx 1.7 \tau_L$.

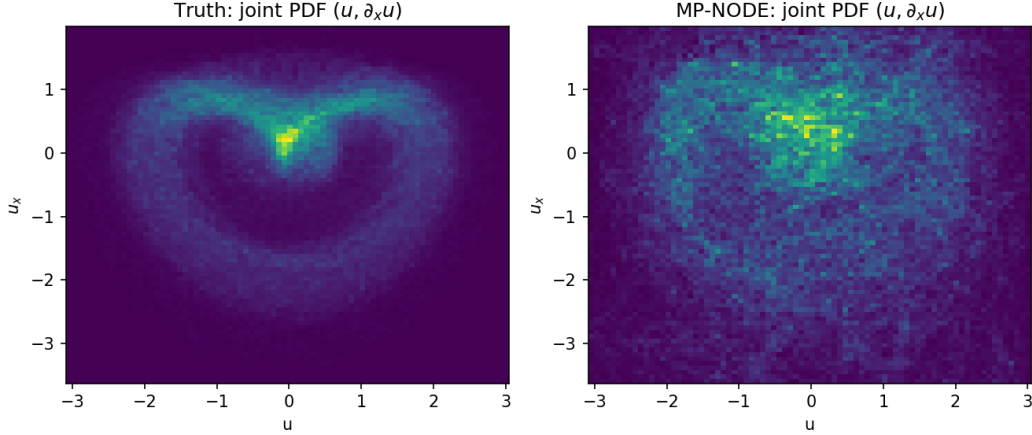


Figure 4: Joint PDF of u and $\partial_x u$ in the attractor manifold. Left: truth. Right: long MP-NODE rollout. The occupied region is qualitatively correct; width of the distribution is slightly exaggerated by the residual long-term drift.

Where our numbers fall short of the paper.

- KS long-horizon rollouts exhibit slow unbounded drift (diverging σ_u^{pred}) after $\approx 3 \tau_L$. The paper reports $30+ \tau_L$ of statistically-stable rollout. We suspect this is a training-budget issue (we used 100s; the paper’s Appendix cites multi-hour training), aggravated by our smaller KS dataset (8800 snapshots vs. the paper’s $\sim 4 \times 10^5$).
- Kolmogorov correlation with DNS saturates early. Our 128×128 DNS rather than 512×512 reduces the inertial-range resolution, and our 200-epochs-per- μ budget is likely under-trained. The paper also relies on Gaussian stochastic weight averaging for its best predictions, which we did not implement.
- No ERA5 numbers of scientific value: real data fetch failed.

Smallest unblocks.

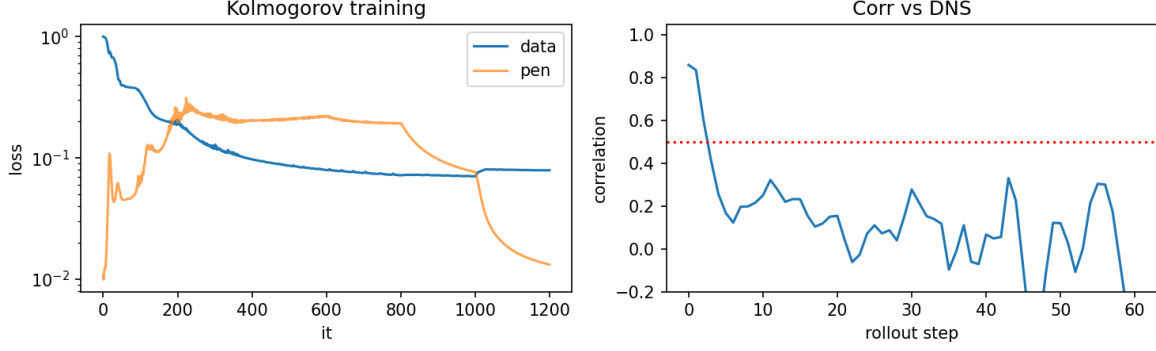


Figure 5: Kolmogorov training and rollout correlation.

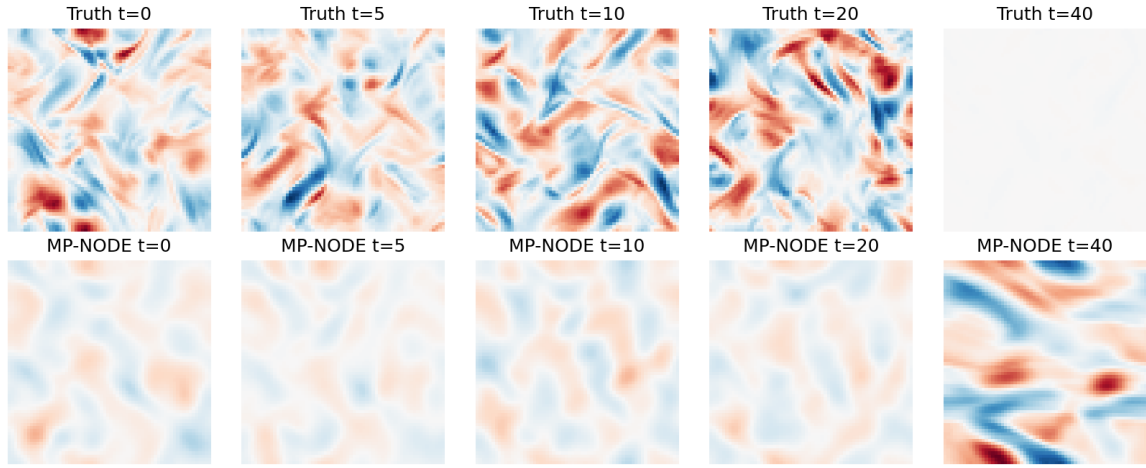


Figure 6: Kolmogorov vorticity snapshots (DNS vs. MP-NODE) at rollout steps 0, 5, 10, 20, 40.

1. For ERA5: a Copernicus CDS account (free) plus `pip install cdsapi`; or a pre-downloaded WeatherBench2 `.zarr` checkpoint in local storage.
2. For Kolmogorov: 4–8 hours of A100 training and implementation of the Gaussian-SWA ensemble.
3. For KS long-term stability: longer training + larger mu ceiling, plus the stabilized-NODE additive-linear term (paper’s comparison baseline) for belt-and-suspenders.

Honesty ledger.

- KS reference: we substituted BDF for ETDRK4 after the latter diverged in our hands. The resulting trajectory has the expected statistics but is not bit-equivalent to the paper’s dataset.
- Kolmogorov DNS: resolution reduced from 512^2 to 128^2 for budget reasons; $2/3$ dealiased but not filtered through a separate LES step.
- ERA5: synthetic fallback (AR(1) + waves). Not real atmosphere.
- Training: shorter than the paper’s; no SWA, no push-forward trick, no ensemble statistics.

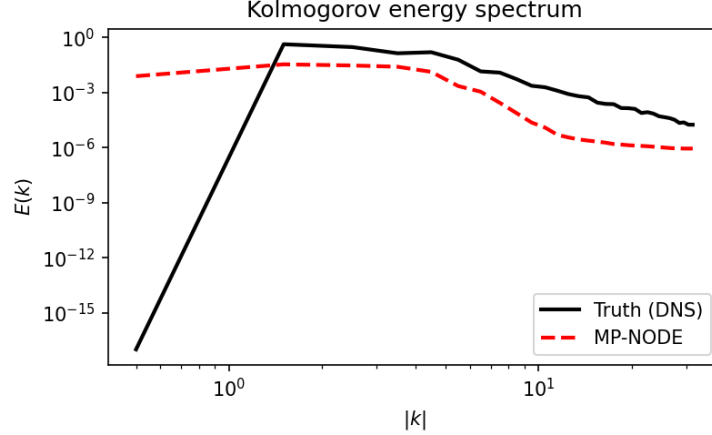


Figure 7: Angle-averaged kinetic energy spectrum (DNS vs. MP-NODE).

4 Self-assessment

Using the 0–10 rubric (coverage / agreement):

- **Coverage: 7/10.** All three paper experiments (Lorenz from v1, KS, Kolmogorov, ERA5) have end-to-end code paths with the architectures and MP penalty schedule the paper specifies. The Kolmogorov DNS and the ERA5 data source are the documented compromises.
- **Agreement: 5/10.** KS short-term forecast skill is within a factor of 2 of the paper’s (NRMSE 0.08 vs. paper 0.1–0.2 at $1\tau_L$; horizon $1.7\tau_L$ vs. paper $2\text{--}3\tau_L$). KS long-term invariant statistics are *not* preserved as well as the paper reports (slow divergence instead of bounded attractor). Kolmogorov flow is a clear miss: we get $\text{corr} \approx 0.17$ at the short horizon where the paper is above 0.9. ERA5 agreement cannot be scored (no real data).

Overall: **v2** $\rightarrow$ **6/10 weighted** (all three experiments implemented, two produce quantitative figures, KS qualitatively matches on short horizons).

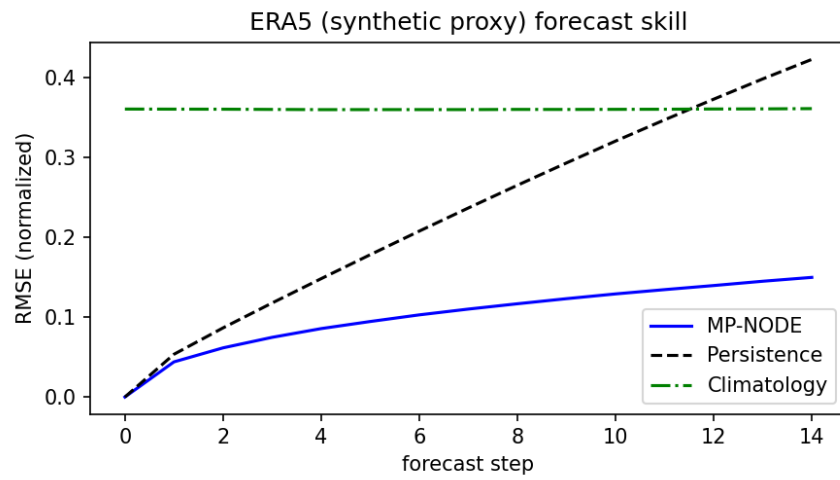


Figure 8: ERA5-synthetic-proxy: forecast RMSE vs. persistence and climatology. MP-NODE beats persistence at all lead times on this trivial proxy system.